# Therapeutic Efficacy and Safety of Recombinant Botulinum Toxin Type A for Moderate to Severe Glabellar Lines: A Randomized, Controlled, Double-blind, Phase III Trial in Chinese Adults

Yan Wu, MD; Xiao Long, MD; Nanze Yu, MD; Jin Chen, MD; Jianyun Lu, MD; Lina Tan, MD; Wei Lai, MD; Li Li, MD; Sufan Wu, MD; Liming Wu, MD; Jiaming Sun, MD; Hongsen Bi, MD; Songmei Geng, MD; Huiping Wang, MD; Shengkang Luo, MD; Zhiqi Hu, MD; Jianmin Yang, MD; Baoxi Wang, MD; Wu Yang, PhD; Xiaoli Sun, MS; and Xiaoyun Sun, MD, PhD

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## Abstract

**Background:** Conventional botulinum toxin type A (BoNT/A) products include non-therapeutic accessory proteins vary in composition and manufacturing processes, which may contribute to differences in purity and biochemical characteristics. YY001 (Retoxin®) is a novel recombinant BoNT/A manufactured via *E. coli* expression to reduce impurities, improve purity, and minimize immunogenic risk.

**Objectives:** To evaluate the efficacy, safety, and immunogenicity of YY001, a recombinant botulinum toxin type A (Chongqing Claruvis Pharmaceutical Co., Ltd., Chongqing, China), in the treatment of moderate-to-severe glabellar lines.

**Methods:** In this multicenter, randomized, double-blind, vehicle- and active-controlled Phase III trial, 529 adults aged 18-65 with moderate to severe glabellar lines were randomized 4:2:1 to receive a single 20 U injection of YY001, Botox®(Allergan Pharmaceuticals Ireland, Westport, Ireland), or vehicle. The primary endpoint was composite response rate (CRR) at Week 4, defined as a ≥2-grade improvement of glabellar line severity assessed by both investigator and participant. Secondary endpoints included onset time, duration, satisfaction, and safety.

**Results:** YY001 achieved a CRR of 86.7% at Week 4, significantly outperforming Botox® (66.7%, $P < .0001$) and vehicle (1.3%, $P < .0001$). It showed faster onset, longer median duration (97 days), and high participant satisfaction. Adverse events were mild and comparable across groups. Only one case of anti-drug antibody was observed; no neutralizing antibodies were detected. A 12-week follow-up and ethnically homogenous population may limit generalizability.

Dr Y. Wu is a professor of dermatology and venerology, Peking University First Hospital, Beijing, China. Dr Long is a professor of plastic surgery and Dr Yu is a associate chief physician of plastic surgery, Peking Union Medical College Hospital (Dongdan campus), Beijing, China. Dr Chen is a professor of dermatology, The First Affiliated Hospital of Chongqing Medical University, Chongqing, China. Dr Lu is a professor of dermatology and venerology and Dr Tan is an assistant professor of dermatology and venerology, The Third Xiangya Hospital of Central South University, Changsha, Hunan, China. Dr Lai is a professor of dermatology and venerology, The Third Affiliated Hospital, Sun Yat.Sen University, Guangzhou, China. Dr Li is a professor of dermatology and venerology, West China Hospital of Sichuan University, Chengdu, Sichuan, China. Dr S. Wu is a professor of plastic and reconstructive surgery, Zhejiang Provincial People's Hospital, Hangzhou, Zhejiang, China. Dr L. Wu is a professor of dermatology and venerology, Affiliated Hangzhou First People's Hospital, Hangzhou, Zhejiang, China. Dr J. Sun is a professor of plastic surgery, Union Hospital Tongji Medical College Huazhong University of Science and Technology, Wuhan, Hubei, China. Dr Bi is an associate professor of plastic surgery, Peking University Third Hospital, Beijing, China. Dr Geng is a professor of dermatology, The Second Affiliated Hospital Of Xi'an Jiaotong University (Xibei Hospital), Xi'an, Shaanxi, China. Dr H. Wang is a professor of dermatology, Tianjin Medical University General Hospital, Tianjin, China. Dr Luo is a professor of plastic and reconstructive surgery, Guangdong Second Provincial General Hoshtal, Guangzhou, Guangdong, China. Dr Hu is a professor of dermatology, Nanfang Hospital, Guangzhou, Guangdong, China. Dr J. Yang is a professor of plastic and reconstructive surgery, Beijing Tsinghua Changgung Hospital, Beijing, China. Dr B. Wang is a professor of dermatology, Plastic Surgery Hospital, Chinese Academy of Medical Sciences, Beijing, China. Dr W. Yang is a co-founder, Xiaoli Sun is a medical manager, and Dr Xiaoyun Sun is the vice president, Chongqing Claruvis Pharmaceutical Co., Ltd., Chongqing, China.

**Corresponding Author:**
Dr Xiaoyun Sun, Chongqing Claruvis Pharmaceutical Co., Ltd., No.28, Yuefu Avenue, DaDi (International) Life Science Park, A2, Beibei District, Chongqing 400700, China.
E-mail: sun.xiaoyun@claruvis.com

**Conclusions:** YY001 demonstrated strong efficacy and a favorable safety profile than Botox®. Longer-term and multicenter studies are needed to assess durability, immunogenicity, and applicability across broader populations.

**Level of Evidence: 1 (Therapeutic)**

Glabellar lines are a type of expression wrinkles that appear between the eyebrows. They are considered dynamic when visible during facial movement and static when they remain apparent at rest. They can significantly affect perceived age and facial expression, fueling demand for effective treatments, and can significantly affect perceived age and facial expression, fueling demand for effective treatments. Botulinum toxin, particularly type A (BoNT/A), is widely used to treat glabellar lines. The U.S. Food and Drug Administration (FDA) approved botulinum toxin type A for clinical use in 1989, followed by its approval for treating glabellar lines in 2002. Currently, the botulinum toxin products approved in China for the glabellar line indication include Botox®(Allergan Pharmaceuticals Ireland, Westport, Ireland), Dysport®(Ipsen Biopharm Ltd. Wrexham, LL13 9UF, UK), Xeomin®(Merz Pharmaceuticals GmbH, Eckenheimer Landstrasse 100, Frankfurt Germany), Letybo®(Hugel, Inc., 61-20 Sinbuk-ro, Sinbuk-eup, Chuncheon, Korea), Hengli® (Lanzhou Biological Products Institute, Lanzhou, China), and Daxxify® (Revance Therapeutics, Inc. Newark, CA).

These approved toxin products are produced via fermentation of *Clostridium botulinum* and often contain accessory non-neurotoxin proteins exposing to patients at varying levels.[1,2] Moreover, fundamental production challenges were identified: both complexing-protein-containing toxins (eg, Botox®) and purified neurotoxins (Xeomin®, Daxxify®) exhibit inherent batch-to-batch variability during *Clostridium*-based expression and activation.[3] The variability in bioconversion processes may contribute to heterogeneity in their biochemical composition and bioactivity. However, these mechanistic considerations remain theoretical and have not been evaluated in the previous clinical trial.[4]

Currently, Chongqing Claruvis Pharmaceutical Co., Ltd has developed YY001 (recombinant botulinum toxin protein type A, brand name: Retoxin®) using genetic engineering technology. With a molecular weight of 150 kDa, YY001 is produced via an *Escherichia coli* expression system, yielding a protein with the same amino acid sequence as natural botulinum neurotoxin type A. The toxin is activated through an in vitro enzymatic activation process, producing a recombinant protein with equivalent biological activity. This approach removes the need for pathogenic *Clostridium* culture and yields a highly purified single-component neurotoxin, although product heterogeneity and comparative immunogenicity relative to onabotulinumtoxinA have not been fully studied. A multicenter, randomized, double-blind, vehicle- and active-controlled Phase III clinical trial (YY001-001-Ⅲ-REFINE) was conducted in China to evaluate the efficacy and safety of a single 20 U dose of YY001 compared with both vehicle and Botox® in treating moderate to severe glabellar lines in Chinese subjects.

# METHODS

## Overall Design

This study was a multicenter, randomized, double-blind clinical trial with 2 parallel control arms, including a vehicle and an active comparator. It aimed to evaluate the efficacy and safety of a single administration of YY001 for the treatment of moderate to severe glabellar lines. The study period consisted of a screening phase (up to 14 days), a treatment phase (single administration), and a follow-up phase lasting 12 weeks. The trial was conducted at multiple investigational sites across China in strict adherence to the Chinese Guidelines for Good Clinical Practice (National Medical Products Administration, 2020) and the International Council for Harmonization Good Clinical Practice (ICH-GCP) E6 R2.[5,6]

Eligible participants were randomly assigned to 1 of 3 groups, respectively in a 4:2:1 ratio: the YY001 group (YY001, planned: 300 participants), the active control group (Botox®, planned: 150 participants), or the vehicle group (planned: 75 participants). Adults aged 18-65 years with moderate to severe glabellar lines (according to a 4-point Facial Wrinkle Scale from 0 to 3 [0 = no line, 1 = mild, 2 = moderate, and 3 = severe]) were enrolled. Key exclusions included prior botulinum toxin use within 6 months, relevant facial procedures, neuromuscular disorders, and contraindicated medications.

Participants were enrolled by site investigators. Randomization was performed centrally using a block randomization schedule generated by an independent statistician and implemented via an interactive web response system (IWRS). Treatment allocation was concealed from investigators, subjects, outcome assessors, sponsor personnel, and statistical teams. To ensure blinding integrity, drug preparation was performed exclusively by authorized unblinded staff. The prepared syringes containing the investigational product, which appeared identical, were handed over to the investigator for administration. Additionally, only unblinded drug preparers—who were not involved in efficacy and safety assessments—had access to allocation codes. These unblinded personnel were not involved in any efficacy assessment.

The study was conducted from December 2023 to May 2024. All participants provided written informed consent prior to enrollment. The study was approved by Peking University First Hospital Ethics Committee (Beijing, China) with the approval number #2023118-002 and complied with the Declaration of Helsinki and ICH-GCP guidelines. Written informed consent for study participation was obtained from all participants. The trial was completed after all randomized participants had attended the Week-12 visit in accordance with the protocol-defined study duration. The final study protocol is available from the corresponding author upon reasonable request.

## Intervention

Each participant underwent a standardized 5-point injection regimen in the glabellar region. The injection sites comprised 2 points per corrugator supercilii muscle bilaterally and 1 point into the procerus muscle. Each injection site received a volume of 0.05 mL, resulting in a total dose of 20 U each for the YY001 and the active control (Botox®) groups, and 0 U for the vehicle group.

All products were reconstituted using identical procedures with 1.25 mL preservative-free 0.9% saline and were used as single-use preparations within 4 hours of reconstitution, following the standardized aseptic handling procedures. Injections were administered intramuscularly using a 30-gauge needle. Syringe appearance and labeling were standardized to preserve the blind.

## Outcome Measures

The primary endpoint was the composite response rate (CRR) at Week 4 post-dose visit, defined as the proportion of participants achieving a

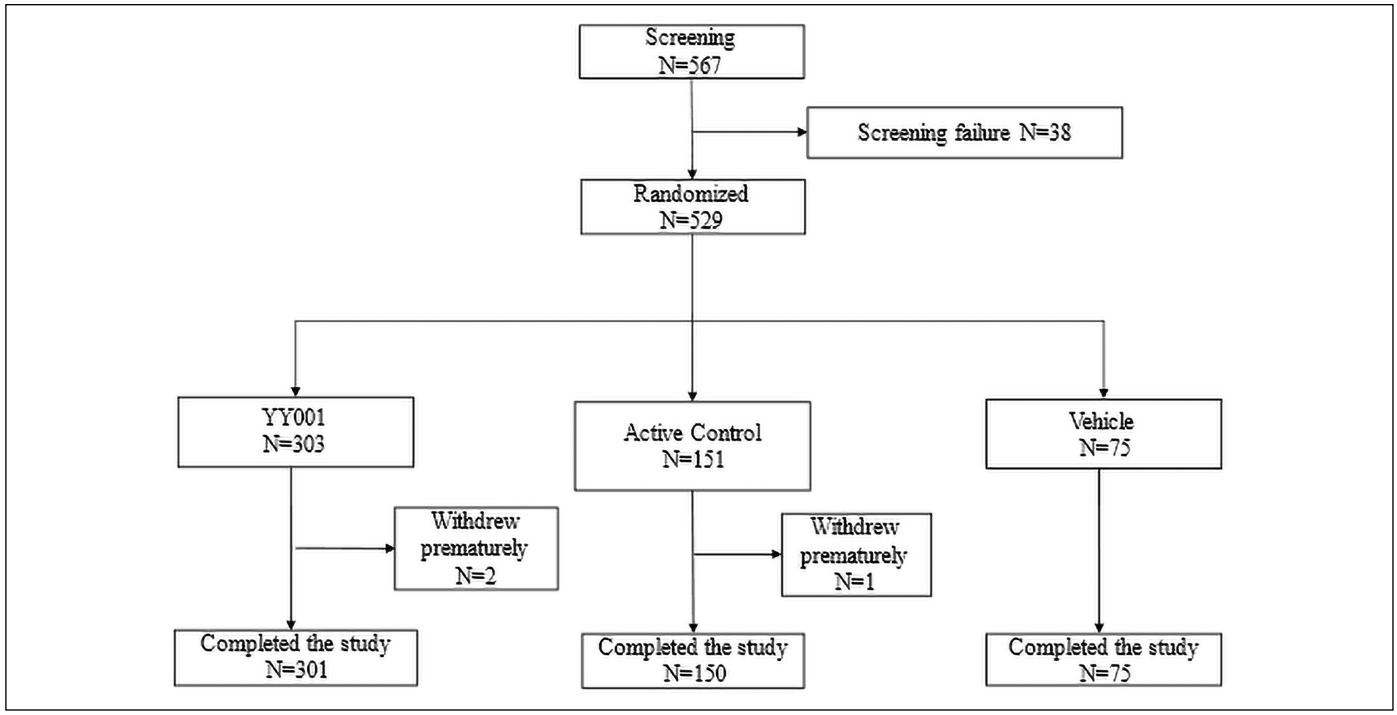



**Figure 1.** CONSORT-compliant participant flow diagram.

glabellar line severity score of 0 or 1 and an improvement of at least 2 grades from baseline, based on both investigator and subject assessments during on-site visits. The severity of glabellar lines at maximum frown was evaluated according to a 4-point Facial Wrinkle Scale from 0-3 (0 = no line; 1 = mild; 2 = moderate; and 3 = severe).

Secondary endpoints included CRR at Weeks 1, 8, and 12; the proportion of participants achieving a glabellar line severity score of 0 or 1, an improvement of at least 2 grades from baseline evaluated by investigator and subject on-site assessments, and an independent review committee (IRC) using photographic assessments at Weeks 1, 4, 8, and 12; participant satisfaction measured via a 7-point scale (1 = Very dissatisfied; 2 = Dissatisfied; 3 = Somewhat dissatisfied; 4 = Indifferent; 5 = Somewhat satisfied; 6 = Satisfied; and 7 = Very satisfied) at Weeks 1, 4, 8, and 12; and the time to onset and duration of effect based on the patient diary records.

Safety endpoints encompassed the incidence of treatment-emergent adverse events (TEAEs), serious adverse events (SAEs), and injection-site reactions. Immunogenicity was tested and assessed exclusively in the YY001 group via anti-drug antibody (ADA) and neutralizing antibody (NAb) testing. Because immunogenicity testing was performed only in the YY001 arm, comparative immunogenicity vs onabotulinumtoxinA was not assessed.

## Statistical Analysis

Efficacy analyses were conducted on the randomized set (RS). Safety analyses included all participants who received at least one dose of the study drug (safety set, SS). The per-protocol set (PP) was used for sensitivity analyses.

Sample-size calculations were derived from assumptions prespecified in the Statistical Analysis Plan. For the primary comparison of YY001 vs vehicle, the expected CRR was 60% for YY001 and ≤10% for vehicle, providing >99% power to detect a between-group difference at a 2-sided $\alpha$ of 0.05. For the non-inferiority comparison of YY001 vs onabotulinumtoxinA, an absolute non-inferiority margin of −10% was selected based on historical effect sizes. Planned sample sizes of 300 participants for YY001 and 150 for the active comparator afforded >90% power for the non-inferiority hypothesis at $\alpha = 0.05$. No changes to the statistical hypotheses or endpoints occurred after trial initiation.

## Primary Endpoint Analysis

The CRR at Week 4 was compared between groups using a Pearson Chi-square test or Fisher's exact test, as appropriate. Ninety-five percent confidence intervals (CIs) for the between-group differences were also calculated.

## Non-Inferiority and Superiority Testing

A −10% non-inferiority margin was used to compare YY001 with active control group (Botox®). If non-inferiority was confirmed, superiority testing was then performed. This gatekeeping strategy-controlled multiplicity across primary comparisons.

## Survival Analysis

Kaplan–Meier methods were applied to analyze time-to-event data, such as onset time and duration of effect. Medians and 95% CIs were derived using the Brookmeyer–Crowley method. Participants without onset or without return to baseline were censored at the last evaluable diary entry. Diary gaps were handled per protocol by carrying forward the last non-missing entry.

## Safety Analysis

Adverse events (AEs) and serious adverse events (SAEs) were coded using MedDRA version 27.0, and severity was assessed by Common Terminology Criteria for Adverse Events (CTCAE) version 5.0, using a grading (severity) scale with a range of grades from 1 (mild) to 5

**Table 1.** Baseline Characteristics of Demographics and Glabellar Line Severity at Maximum Frown

| | YY001 | Active control | Vehicle |
|---|---|---|---|
| | N = 301 | N = 150 | N = 75 |
| Gender: male n (%) | 40 (13.3) | 24 (16.0) | 11 (14.7) |
| Age (y): mean (SD) | 35.3 (8.9) | 36.2 (9.1) | 35.4 (8.3) |
| Race: Han n (%) | 286 (95.0) | 141 (94.0) | 73 (97.3) |
| BMI (kg/m$^2$): mean (SD) | 22.33 (3.21) | 22.48 (3.29) | 22.57 (3.56) |
| Glabellar line severity at maximum frown | | | |
| Assessed by Investigator n (%) | | | |
| 3 (Severe) | 100 (33.2) | 54 (36.0) | 27 (36.0) |
| 2 (Moderate) | 201 (66.8) | 96 (64.0) | 48 (64.0) |
| Assessed by participant n (%) | | | |
| 3 (Severe) | 101 (33.6) | 57 (38.0) | 23 (30.7) |
| 2 (Moderate) | 200 (66.4) | 93 (62.0) | 52 (69.3) |

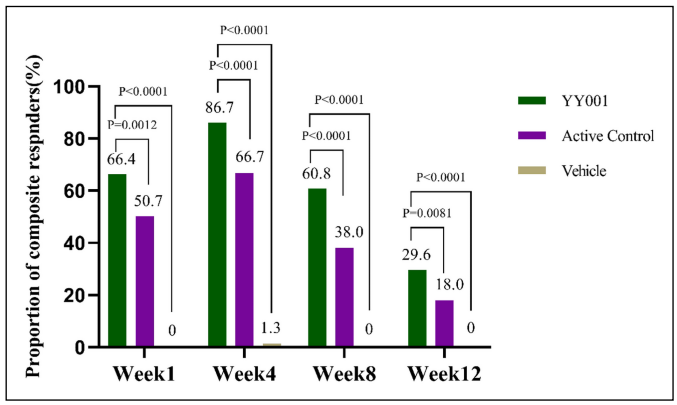



**Figure 2.** Composite response rates (CRRs) at Week 1, Week 8, and Week 12, in which the CRR is defined as the proportion of participants achieving a glabellar line severity score of 0 or 1 and an improvement of at least 2 grades from baseline, based on both investigator and subject assessments during on-site visits.

(death). All statistical analyses were prespecified in the Statistical Analysis Plan (SAP) and conducted using validated statistical software (SAS version 9.4).

# RESULTS

## Participant Flow and Baseline Characteristics

A CONSORT-compliant participant flow diagram summarizing screening, randomization, treatment allocation, and study completion is provided in Figure 1. A total of 567 participants were screened. Thirty-eight participants (6.7%) failed the screening. Among those, 32 participants (5.6%) did not meet the inclusion criteria, and 6 participants (1.1%) withdrew their informed consent. Ultimately, 529 (93.3%) participants were randomly assigned to groups: 303 (53.4%) in the YY001 group; 151 (26.6%) in the active control group; and 75 (13.2%) in the vehicle group. Of the 529 participants, 526 participants (99.4%) completed the study, with 3 participants having withdrawn prematurely from the study (2 from the YY001 group and 1 from the active control group).

For baseline characteristics, all participants were of Asian descent, with a balanced distribution of age (35.3Y vs 36.2Y vs 35.4Y), gender (Male: 13.3% vs 16.0% vs 14.7%), and BMI (22.33 kg/m$^2$ vs 22.48 kg/m$^2$ vs 22.57 kg/m$^2$) across different cohorts. No significant differences in baseline characteristics, including glabellar line severity at maximum frown, were observed among the 3 treatment arms (Table 1). Except for 3 subjects who withdrew prematurely (including one in the YY001 group who did not receive the injection, and one each in the YY001 group and the active control group who withdrew after injection), please refer to the subject flow diagram (Figure 1). The rest participants completed the 12-week follow-up period; and all study visits occurred as scheduled (the follow-up duration was 12 weeks for all participants, with no variation in scheduled study visits). Three anonymized patient vignettes (2 treated with YY001 and one receiving Botox® as active control) are provided as examples in the Appendix, available online at https://doi.org/10.1093/asj/sjag042.

## Efficacy

### *Efficacy Outcomes*

At Week 4 post-dose, the CRR was 86.7% for YY001, 66.7% for Botox®, and 1.3% for vehicle. YY001 demonstrated superiority vs vehicle, and met the prespecified criteria for non-inferiority followed by superiority vs onabotulinumtoxinA, in accordance with the hierarchical testing strategy. Sensitivity analyses confirmed the robustness of these results.

The response rates for YY001 at earlier and later time points were as follows: CRRs were 66.4% at Week 1; 60.8% at Week 8; and 29.6% at Week 12. In comparison, the active control group (Botox®) showed CRRs of 50.7%, 38.0%, and 18.0%, respectively, while the vehicle group exhibited 0% response at all evaluated time points (Figure 2). Across all time points, the magnitude and pattern of response favored YY001 over both comparator groups.

### *Secondary Efficacy Outcomes*

The proportion of participants achieving a score of 0 (none) or 1 (mild) for glabellar lines at maximum frown was 90.7% at Week 1, and 95.0% at Week 4. This proportion declined to 88.4%

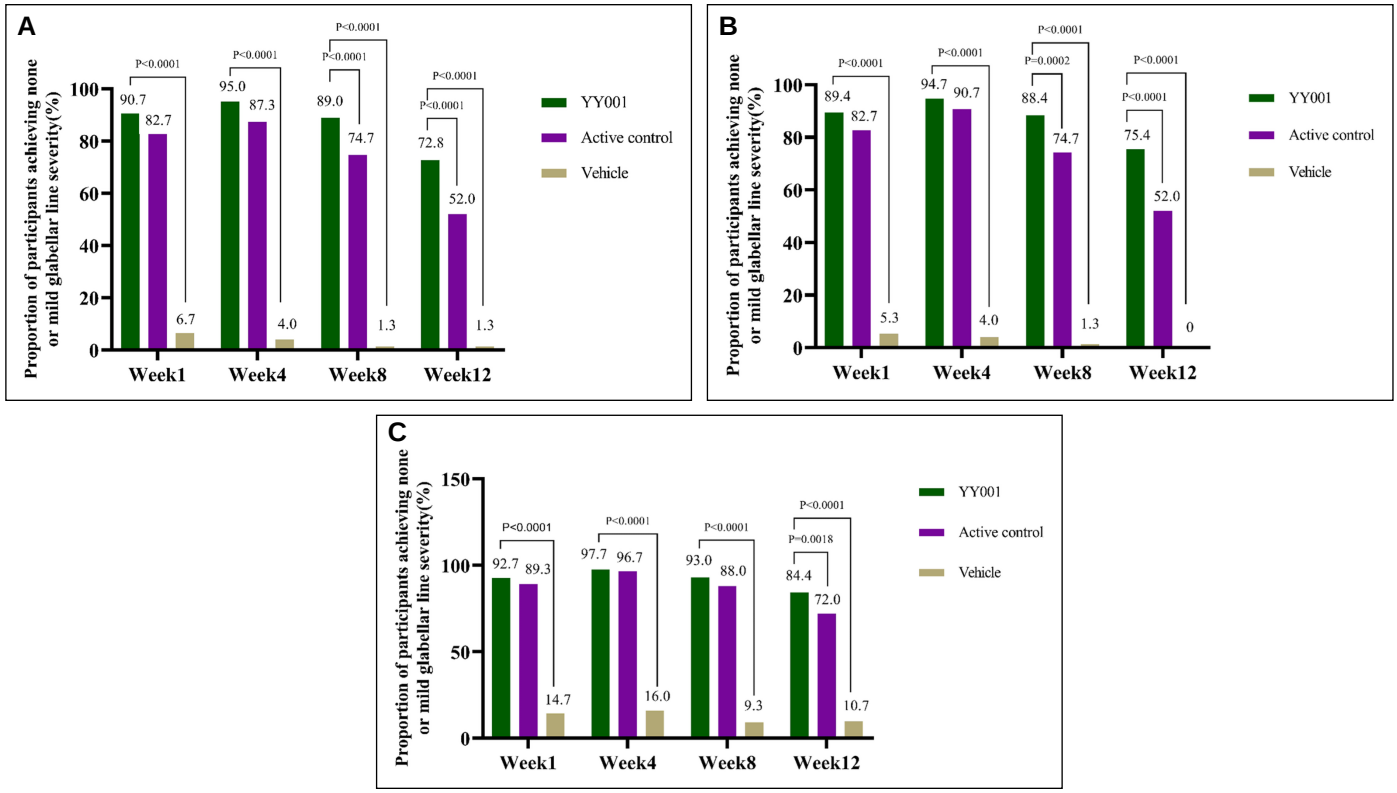



**Figure 3.** (A) Proportion of participants who achieved a score of 0 or 1 of glabellar line severity at maximum frown on the investigators' assessments at Week 1, Week 4, Week 8, and Week 12; (B) Proportion of participants who achieved a score of 0 or 1 of glabellar line severity at maximum frown on the participants' assessments at Week 1, Week 4, Week 8, and Week 12; (C) Proportion of participants who achieved a score of 0 or 1 of glabellar line severity at maximum frown on the photographic assessments conducted by the Independent Review Committee (IRC) at Week 1, Week 4, Week 8, and Week 12.

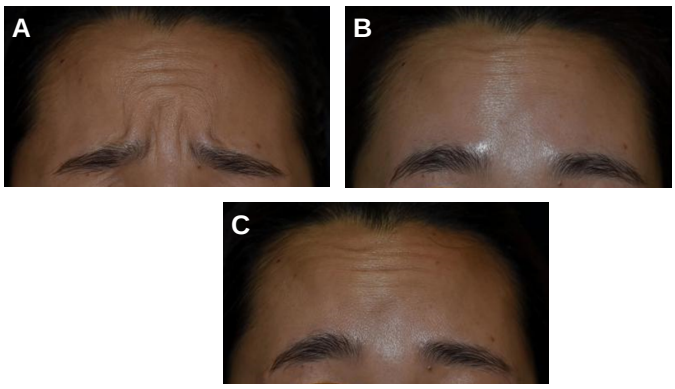



**Figure 4.** Clinical photographs of a 33-year-old Chinese female participant treated with YY001. The participants had a baseline Investigator's and Subject's FWS score of 3 and achieved a score of 0 at both Weeks 4 and 12 post-treatment. (A) Baseline; (B) Week 4 post-treatment; (C) Week 12 post-treatment.




**Figure 5.** Clinical photographs of a 25-year-old Chinese female participant treated with YY001. The participants had a baseline Investigator's and Subject's FWS score of 3 and achieved a score of 0 at both Weeks 4 and 12 post-treatment. (A) Baseline; (B) Week 4 post-treatment; (C) Week 12 post-treatment.

by Week 8 and 75.4% at Week 12, yet remained substantially higher than rates observed in both the Botox® and vehicle control groups throughout the observation period (Figure 3A).

Participants' self-assessment and photographic assessments conducted by the Independent Review Committee (IRC) further corroborated these findings. Across all post-treatment visits (Weeks 1, 4, 8, and 12) (Figure 3B and C), the proportion based on participant assessments and photographic evaluation paralleled onsite investigator assessments, demonstrating good consistency across evaluation modalities. Representative photographs of the 3 composite responders are shown before and after treatment in Figures 4-6.

Analysis of treatment onset within the first 7 days post-injection revealed a rapid response for YY001, with the earliest recorded onset occurring on Day 1.The median onset time for YY001 was 3 days, comparable to Botox® (Supplemental Table 1, available online at https://doi.org/10.1093/asj/sjag042), indicating a prompt clinical benefit.

Regarding the duration of effect, participants treated with YY001 sustained at least a one-grade improvement for a median of 97 days as assessed by investigators (see Supplemental Figure 1, available online at https://doi.org/10.1093/asj/sjag042) and over 91 days as reported by participants (see Supplemental Figure 2, available online at https://doi.org/10.1093/asj/sjag042). This duration surpassed that observed in

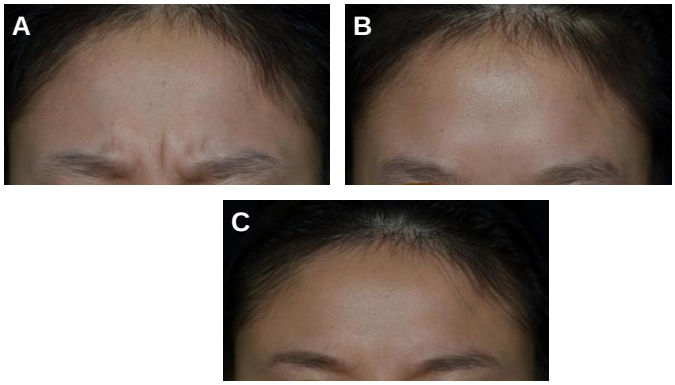



**Figure 6.** Clinical photographs of a 21-year-old Chinese female participant treated with YY001. The participants had a baseline Investigator's and Subject's FWS score of 3 and achieved a score of 0 at both Weeks 4 and 12 post-treatment. (A) Baseline; (B) Week 4 post-treatment; (C) Week 12 post-treatment.

the active comparator (median 90 days) and vehicle groups (median 55-57 days). These findings indicate that YY001 produced a rapid onset and maintained improvement over the 12-week observation window, consistent with findings from the IRC based on photographic assessments at each post-treatment visit (see Supplemental Figure 3, available online at https://doi.org/10.1093/asj/sjag042).

Patient satisfaction scores aligned with clinical outcomes. The proportion of participants reporting high satisfaction (score ≥6 on a 7-point scale) was consistently higher in the YY001 group compared to both comparators at all time points. Satisfaction remained above 70% at Week 4 and above 50% at Week 8 in the YY001 group (see Supplemental Figure 4, available online at https://doi.org/10.1093/asj/sjag042), highlighting the meaningful and lasting patient-perceived benefits.

## Safety and Tolerability

The overall incidence of AEs was similar across the 3 groups, although the incidence of treatment-related AEs in the YY001 group (4.6%, 14/302) was lower than that in the active control group (9.3%, 14/150) and comparable to the vehicle (2.7%, 2/75) (Table 2). The AEs of CTCAE grade 3 or higher were respiratory tract infection, mixed hemorrhoids, left otitis media (acute, non-supp), elevated blood creatine kinase, periapical periodontitis, bronchial pneumonia and tooth defect. All treatment-related AEs were mild or moderate (CTCAE grade 1 or grade 2) in severity. The incidence of injection site reactions (ISRs) was 2.6% (8/302) in the YY001 group, 2.7% (2/75) in the vehicle group, and 8.0% (12/150) in the active control group, respectively. All injection site reactions were mild (CTCAE grade 1) and most resolved on their own within one week. Regarding ocular AEs, eyelid edema incidence was 0% and eyelid ptosis occurred in 0.7% (2/302) of participants with YY001.

## Immunogenicity

The test and assessment of immunogenicity was confined to the YY001 cohort, among which only one subject developed treatment-emergent anti-drug antibodies (ADA). This drug-related ADA-positive case showed negative baseline results, seroconverting to positive at both 4 and 12 weeks post-dose, and the participant achieved the composite response level at Week 4 post-treatment without any reported AEs. No participants tested positive for neutralizing antibodies. Because immunogenicity testing was performed only in the YY001 arm, comparative immunogenicity vs onabotulinumtoxinA was not assessed.

**Table 2.** Summary of TEAEs During 12-Week Treatment Period (SS)

| | YY001 (N = 302) | Active control (N = 150) | Vehicle (N = 75) |
|---|---|---|---|
| All TEAE | 104 (34.4) | 61 (40.7) | 32 (42.7) |
| Treatment related[1] | 14 (4.6) | 14 (9.3) | 2 (2.7) |
| TEAE CTCAE ≥grade 3 | 4 (1.3) | 3 (2.0) | 0 (0) |
| Treatment related and CTCAE ≥grade 3 | 0 (0) | 0 (0) | 0 (0) |
| SAE | 1 (0.3) | 2 (1.3) | 0 (0) |
| Treatment related | 0 (0) | 0 (0) | 0 (0) |
| AE led to discontinuation | 0 (0) | 0 (0) | 0 (0) |
| Treatment related and led to discontinuation | 0 (0) | 0 (0) | 0 (0) |
| Most common TEAEs (≥2%, in YY001 group) by PT | | | |
| Upper respiratory tract infection | 12 (4.0) | 13 (8.7) | 7 (9.3) |
| Urinary tract infection | 8 (2.6) | 2 (1.3) | 3 (4.0) |
| Cough | 6 (2.0) | 2 (1.3) | 0 (0) |
| General Disorders and Administration Site Conditions | 8 (2.6) | 12 (8.0) | 2 (2.7) |
| Swelling | 7 (2.3) | 8 (5.3) | 1 (1.3) |
| Pain | 1 (0.3) | 3 (2.0) | 1 (1.3) |
| Erythema | 0 (0) | 2 (1.3) | 0 (0) |
| Pruritus | 1 (0.3) | 1 (0.7) | 0 (0) |
| Discomfort | 1 (0.3) | 0 (0) | 0 (0) |
| Hypoaesthesia | 0 (0) | 1 (0.7) | 0 (0) |
| Induration | 0 (0) | 1 (0.7) | 0 (0) |
| Bruising | 0 (0) | 1 (0.7) | 0 (0) |

Adverse events are defined as those assessed as "definitely related," "probably related," or "possibly related" to the study drug. If the causality assessment is missing, the event will conservatively be considered related to the study drug. AE, adverse event; CTCAE, Common Terminology Criteria for Adverse Events; PT, preferred term; SAE, serious adverse event; SOC, system organ class; TEAE, treatment-emergent adverse event.

## DISCUSSION

YY001 is a recombinant botulinum toxin type A produced through an *E. coli* expression system and in vitro activation, producing the toxin with an amino acid sequence identical to native botulinum toxin type A without the need for pathogenic *Clostridium botulinum* cultures. This approach results in a highly purified 150 kDa neurotoxin free from complexing proteins, although the potential clinical implications of the single-component design—particularly regarding immunogenicity—cannot be determined from this study.[1,2,7] Notably, comparative immunogenicity vs onabotulinumtoxinA was not assessed.

In the Phase I/II clinical trial YY001-001, YY001 exhibited a favorable safety and efficacy profile comparable to Botox® 20 U across single-dose administrations of 10, 20, and 30 U, with no emergent safety concerns. Phase I efficacy assessments demonstrated investigator-evaluated responder rates (defined as ≥2-grade improvement in

glabellar line severity from baseline) at Week 4 post-injection of 80% for YY001 20 U, 90% for YY001 30 U, and 70% for Botox® 20 U. Subsequent Phase II results showed responder rates of 83.7% for YY001 20 U vs 73.3% for the active comparator (Botox). Although Phase I data suggested a potential dose–response relationship for YY001 (30U:90% vs 20 U:80%), the absence of statistically superior efficacy at the higher dose in later-stage analysis, coupled with comparable performance between YY001 20 U (83.7%) and Botox® (73.3%)[8] in Phase II, supported the selection of the 20-unit dose as optimal based on benefit-risk assessment for confirmatory trials.

In this Phase III trial, the CRR results demonstrated that YY001 (20 U) demonstrated a significantly higher responder rate than vehicle and met the criteria for non-inferiority followed by superiority vs onabotulinumtoxinA throughout the 12-week evaluation period (CRR: 66.4% at Week 1; 86.7% at Week 4; 60.8% at Week 8; and 29.6% at Week 12). This treatment benefit was consistently observed across investigator, participant, and independent photographic assessments, and was robust across demographic subgroups.

Notably, pivotal Phase III studies of competing neurotoxins reported CRRs at Week 4 of 61% for Daxxify® (40 U) in Chinese subjects, 60% for Xeomin® (20 U), and 60% for Dysport® (50 U).[9-13] Yet, because methodologies, dosing regimens, and assessment criteria vary across trials, such cross-study comparisons should be interpreted cautiously and are not intended to imply comparative efficacy. Additionally, the proportion of subjects achieving a glabellar line severity rating of none or mild (Grade 0/1) as assessed by investigators was 90.7% at Week 1, 95.0% at Week 4, and 72.8% at Week 12 for YY001. These rates were not only statistically superior to the active comparator (82.7%, 87.3%, and 52.0%, respectively) but were also comparable to those reported for competitor products Xeomin® (20 U: 90.5%, 92.5%, 67.2%)[14] and Daxxify® (40U: 93.8%, 97.5%, 86.2% from studies SAKURA 1 and SAKURA 2).[15] Although these proportions were higher than those observed in the active comparator group and generally in line with ranges reported in the literature for other BoNT/A formulations, direct comparisons are limited by methodological heterogeneity. As defined by this trial's design parameters, YY001 demonstrated strong clinical efficacy.

Collectively, this study and cross-trial data establish that YY001 exhibits durability, peak effect, and overall performance comparable to marketed toxin products, while maintaining non-inferior efficacy relative to the active comparator. Safety findings reaffirm the favorable risk profile of YY001. The incidence and severity of treatment-related adverse events (TRAEs) for YY001 were lower than those in active control group (4.6% vs 9.3%). Regarding distant spread of toxin-related effects, the incidence of eyelid ptosis in the YY001group was 0.7%. Historical data show that Xeomin® had an incidence of 0.9% (including 0.2% eyelid ptosis and 0.7% facial paresis/brow ptosis); Daxxify® demonstrated 2% eyelid ptosis and 1% facial paresis; Botox® reported 3% eyelid ptosis; and Dysport® showed 2% eyelid ptosis.[10-13]

Although published rates of ptosis for other BoNT/A products typically range from approximately 0.2% to 3%, such comparisons should be interpreted carefully given differences in dosing, study design, and injection technique. Especially, only one treatment-emergent anti-drug antibody (ADA) case was observed, with no detectable neutralizing antibodies. This ADA-positive subject maintained optimal efficacy and safety profiles. As immunogenicity assessments were conducted only in the YY001 arm, no comparative immunogenicity conclusions can be drawn.

## Limitations

Although the results of this study are compelling, they should be interpreted with caution due to the single-dose regimen, short (12-week) follow-up duration. All participants were of East Asian descent, which may affect the generalizability of efficacy outcomes. Prior literature has suggested anatomical differences such as reduced muscle mass and thinner dermis may enhance responsiveness to botulinum toxins in Asian populations.[16] These differences could partly explain the high response rates observed and highlight the need for replication in more diverse cohorts. Prospectively, an open-label, single-arm, multi-dose, long-term follow-up study (YY001-001-Ⅲ-RENEW, ClinicalTrial.gov No. NCT06583486) is being conducted at 16 sites in China, enrolling 488 participants over 5 treatment cycles with a follow-up of 64 weeks. Additionally, global multicenter studies involving broader demographic populations and more treatment indications have been planned. Hopefully, this study along with the ongoing and future planned studies will examine the efficacy and safety of YY001 in a longer follow-up period and demographically diverse population, thereby help to address these concerns.

## CONCLUSIONS

In conclusion, YY001 demonstrated robust efficacy and a favorable safety profile compared with vehicle and achieved higher Week-4 CRR than onabotulinumtoxinA within the prespecified statistical testing framework in treating moderate to severe glabellar lines. These findings support YY001 being a promising addition to available BoNT/A treatment options, although longer-term and multicenter studies are required to better characterize its durability, immunogenicity, and use across diverse populations.

### Supplemental Material

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Drs W. Yang and Xiaoyun Sun reported being employees and shareholders of Chongqing Claruvis (Chongqing, China). Dr Xiaoli Sun reported being an employee of Chongqing Claruvis (Chongqing, China). The other authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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